

July Edition: AI & Analog Remaining the Bright Spots into Earnings

Jeff Pu, CFA

SFC CE No. BNO719

jeffpu@gfgroup.com.hk

Zhou Yang

SFC CE No. BSF949

zhouyang@gfgroup.com.hk

Alicia Xia

SFC CE No. BSS651

aliciaxia@gfgroup.com.hk

Evan Lee

SFC CE No. BVI629

evanlee@gfgroup.com.hk

Michelle Jing

SFC CE No. BUK594

michellejing@gfgroup.com.hk

Market resilient amid rising tariffs: Following our *June Edition on June 11*, the tech indices have recovered from the early April plunge and many tech stocks reached record highs (i.e., Nvidia, Broadcom, TI, etc), as the AI fueled demand remains robust, and several semi sub-sectors such as analog extended the visibility of recovery. Near term, the market will watch the tariff rate which may rise if threatened tariffs go into effect August 1st as scheduled. Moreover, a 50% tariff on copper and yet-to-be-announced tariff on semiconductors will also raise the effective tariff. Semiconductor tariffs are yet to be announced and it is unclear if the administration will also seek to tariff a broader array of electronics products. That said, we expect the tech shares to be in range-bound and continue to prefer AI and analog sectors. On the other hand, we are cautious on WFE, memory (including DDR4) and Apple names.

Earnings season began, preferring AI: On July 7, FII (601138 CH) announced the stronger than expected 2Q25 profit guidance, echoing our earlier *preview report on Hon Hai (2317 TT Buy) on July 3*. Moreover, ASML is expected to hold earnings call on July 16, and as said in our *ASML (Underperform) report*, we expect the 2Q order to be lower than expected, due to fewer EUVs from TSMC, despite the strong China demand. Following ASML's, TSMC (2330 TT Hold)'s earnings, on July 17, is expected to be an inline one, due to the likely inline 2025 revenue guidance and capex, while the market will watch the inventories, NTD impact and 3Q25 guidance given high base. Nevertheless, we remain positive on AI, backed by intact CSP capex, GB ramp-up, as well as significant visibility on AMD and ASICs. We like AI names such as Nvidia (NVDA Buy), Hon Hai (2317 TT Buy), Broadcom (AVGO Buy), Celestica (CLS Buy). Following our earlier non-consensus Buy call on Marvell (MRVL Buy), we initiated Supermicro (SMCI Buy) on June 24 on the acceleration in Blackwell and GB300. Aside from AI, we continue to like analog sector and expect a beat & raise for TI's earnings.

Following earlier ASIC-fueled rally, mixed views emerging: According to our 1.6T demand report on July 1st, we see an upside from Google's v7, leading to the recent 1.6T upward revision. On the other hand, we continue to see a mild volume for Meta's MTIA in 2026, due to a lack of deployment scale and the late launch of its Iris project. For AWS, market noises emerged, with some market watchers highlighting a cut in 4Q25 for CCL, PCB and some other components. Indeed, AWS order trends are mixed, as we see an upward revision on CoWoS from 85k to 90k in 2025. In sum, we prefer Optical module (COHR, LITE), CCL (EMC) and design services (Broadcom, Marvell, MediaTek).

June monthly sales/data: TSMC revenue -18% MoM/+27% YoY, inline with consensus and our estimates, underpinned by strong AIGPU/ASIC demand. Top 5 ODM NB shipment increase 23% MoM, but PC outlook in 2H remains uncertain. Aspeed resumed +55% YoY growth driven by both AI and general server. Technology sector is generally impacted by FX. Details on page 5.

GFHK Tech Most Read: 1) Yang Zhou on **EMC (2383 TT)** on CCL upgrade trends; 2) Yang Zhou on **AMSL (Underperform)** on 2Q25 preview; 3) Jeff Pu on **Hon Hai (2317 TT)**'s preview; 4) Alicia Xia on **1.6T upside**; 5) Jeff Pu on **Supermicro (SMCI Buy)** initiation.

Figure 1: Cyclical Indicators

	Short-term	Mid-term		Short-term	Mid-term
DRAM	↗	→	iPhone	↗	↘
NAND	→	→	Android	→	→
Analog	↗	↗	PC	→	→
Optical	↗	↗	AI Server	↗	↗

Sources: GF Securities (Hong Kong) Brokerage

Risks

1) Alternative technologies; 2) AI demand and capex slowdown; 3) IT demand slowdown; 4) product delays.

Table of Contents

<u>GF Tech Important Stock Calls</u>	<u>3</u>
<u>Monthly sales and consensus tracking</u>	<u>5</u>
<u>Tariff Implications to Tech</u>	<u>10</u>
<u>Downstream tech – Smartphone & Apple</u>	<u>12</u>
<u>Downstream tech – PC</u>	<u>14</u>
<u>Upstream tech – AI Semi & Supply Chain</u>	<u>16</u>
<u>Upstream tech – Memory & HBM</u>	<u>18</u>
<u>Upstream tech – WFE</u>	<u>22</u>
<u>Upstream tech – Optical</u>	<u>24</u>

GF Tech Important Stock Calls

In the past one month, our key reports and stock calls are:

Yang Zhou on EMC (2383 TT Buy) on share gain and CCL upgrade, reported date on July 11: CCL market continues consolidating toward EMC, which is entering Google's V6 supply chain and appears well-positioned to become the primary supplier for V7's transition to M9 materials. Key M9 developments include: 1) Google's V7, 2) NVIDIA's Rubin for both switch tray and compute tray, and 3) The orthogonal backboard of Rubin Ultra with EMC's 896K3. As ASICs further migrate to N2/A16 nodes in 2027, we expect M9 CCL likely become the industry standard. EMC remain our top pick in the AI driven PCB/CCL upcycle for share gain and content increase and we expect another revenue acceleration in 4Q25 driven by Meta and Google, alongside with new capacity addition in Zhongshan China. That said, we now forecast EMC's revenue to be NT\$94.8/117.5bn in 2025/2026 and our TP NT\$1297 is based on 22x 2026PE.

Yang Zhou on AMSL (Underperform) on 2Q25 preview, reported date on July 7: We forecast 2Q25 order intake at ~€5bn (including ~€2.3bn of EUV booking), below bearish view of ~€6.5-7bn. Though market is turning more positive on TSMC's order on accelerating Arizona fab expansion, we see TSMC's 2026 EUV demand remain in 14-16 units range. Intel remains absent from 2Q25 bookings, with its 2026 EUV demand likely to decline sharply to 3-5 units compared to 10 units in 2025. On the DUV front, orders from China remain robust at ~€2 billion, driven primarily by Huali's capacity expansion. As we still anticipate a broad-based correction in SPE, we maintain our Underperform rating on ASML yet we raise our target price to \$676 based on 22x 2026E P/E.

Jeff Pu on Hon Hai (2317 TT Buy)'s preview, reported date on July 3: Fundamentally, we believe that AI supply chain will remain strong backed by limited impact from tariffs (including Section 232), intact CSP capex and rising GB200/ASIC demand. For Hon Hai, we forecast the GB200+GB300 units to be 14k in 2025, with improving margin profile. Hence, we expect Hon Hai and FII to post strong 2Q25-3Q25 results. For 2026, the GB300 will continue to be the main driver, while the company should lead in VR200 (leading supplier in 5 CSPs) and to ramp ASIC servers. As a result, we keep our earnings estimates largely unchanged and raise TP from NT\$185 to NT\$213, as we roll over our valuation basis to 15x 2025E/2026E P/E.

Alicia Xia on 1.6T upside, reported date on July 1: We update that the total shipment of 1.6T will reach 6mn in 2026, mainly driven by Google's better than expected demand. Shipment of TPU revised up in 2025, driven by better demand and delay of TPUv7, with strong momentum in 2026. Other than TPU, we expect that MTIA v1/v1.5 version to be ~0.5m in 2026, less than market's positive expectation of over 1m. Our optimistic assumption for Meta's 800G optical module is 9m in 2026. Minerva's scale out L1 may use optical module or AEC. If the L1 all use optical module, the ASIC to optical module ratio will be 1:7. Considering GPU and traditional data center upgradation, we expect that the optimistic outlook for Meta's 800G optical modules in total will be 9m.

Jeff Pu on Supermicro (SMCI Buy) initiation: Bottoming Out and Coming Back, reported date on June 24: We initiate coverage on Supermicro (SMCI US) with a Buy rating and TP of \$59. SMCI's share price was volatile in the past two years due to events of: 1) reaching all-time high share price in March 2024 amid AI server demand; 2) the resignation of auditor Ernst & Young in October 2024; 3) hiring new auditor BDO and announcing delayed filings plan in November 2024; 4) filed financials on time, avoiding NASDAQ delisting in February 2025. Moreover, we believe the most overhangs such as internal audit concern, U.S. restriction to Asia, late Blackwell ramp-up, etc have been removed. That said, driven by robust AI demand, SMCI's re-acceleration in Blackwell/GB300 and rising visibility in emerging market, we expect it to be traded up. As such, our TP is based on 18x FY26 P/E, slightly higher than its mid-cycle valuation.

Figure 2: Coverage summary

Company	Ticker	Rating	TP	Mkt Cap	EPS		P/E		ROE (%)		P/B	
				US mn	2025E	2026E	2025E	2026E	2025E	2026E	2025E	2026E
TSMC	2330 TT	Hold	NT\$968	976,918	NT\$ 60.7	NT\$ 71.9	18.1x	15.3x	29.2	27.8	5.3x	4.3x
Nvidia	NVDA US	Buy	US\$153	4,004,040	US\$4.38	US\$5.54	37.5x	29.6x	89.9	88.7	33.7x	26.3x
AMD	AMD US	Buy	US\$139	233,742	US\$4.05	US\$5.25	35.6x	27.5x	10.8	13.1	3.8x	3.6x
MTK	2454 TT	Buy	NT\$1,730	77,889	NT\$ 69.3	NT\$ 75.9	20.5x	18.7x	28.0	28.9	5.7x	5.4x
BYDE	00285 HK	Buy	HK\$45.0	9,702	CNY 2.09	CNY 2.91	16.2x	11.7x	13.6	16.7	2.1x	1.8x
Broadcom	AVGO US	Buy	US\$262	1,295,336	US\$6.47	US\$8.16	42.6x	33.7x	45.1	52.7	18.8x	17.4x
Marvell	MRVL US	Buy	US\$100	63,251	US\$2.81	US\$3.86	26.1x	19.0x	14.6	16.7	3.8x	3.2x
TI	TXN US	Buy	US\$240	199,555	US\$6.00	US\$7.21	36.6x	30.5x	33.1	35.7	12.0x	10.8x
Micron	MU US	Hold	US\$116	137,775	US\$7.80	US\$9.52	15.8x	12.9x	17.8	17.9	2.7x	2.3x
SK Hynix	000660 KS	U/P	W 132,062	155,959	W 43,609	W 42,787	6.8x	6.9x	22.9	15.4	1.6x	1.1x
SanDisk	SNDK US	U/P	US\$31	6,822	US\$3.04	US\$5.12	15.4x	9.2x	3.8	5.4	0.6x	0.5x
ASML	ASML US	U/P	US\$676	315,888	EU€24.80	EU€26.06	27.6x	26.3x	47.1	41.0	13.1x	10.8x
Onto	ONTO US	Buy	US\$127	5,028	US\$5.17	US\$6.34	19.9x	16.2x	12.4	12.5	2.6x	2.5x
ASMPT	00522 HK	Buy	US\$64	3,242	US\$1.43	US\$2.89	41.0x	20.3x	6.5	8.9	1.6x	1.4x
Apple	AAPL US	Hold	US\$200	3,172,519	US\$7.19	US\$7.69	29.6x	27.6x	237.5	487.5	95.1x	250.0x
Hon Hai	2317 TT	Buy	US\$213	71,150	US\$13.11	US\$15.35	12.3x	10.5x	10.5	11.5	1.3x	1.2x
Dell	DELL US	Hold	US\$121	86,832	US\$9.31	US\$10.26	13.7x	12.5x	N/A	N/A	N/A	N/A
Super Micro	SMCI US	Buy	US\$59	30,056	US\$3.26	US\$3.92	15.4x	12.8x	23.5	22.0	3.6x	2.8x
Lenovo	00992 HK	Buy	HK\$11	15,597	HK\$0.99	HK\$1.29	9.6x	7.4x	24.8	26.7	20.7x	17.3x
Xiaomi	01810 HK	Buy	HK\$64.6	191,260	CNY 1.76	CNY 2.20	32.3x	25.9x	19.0	19.2	6.1x	5.0x
EMC	2383 TT	Buy	NT\$1,297	11,175	NT\$ 44.0	NT\$ 59.0	21.4x	16.0x	30.8	24.4	6.6x	3.9x
KBLaminates	01888 HK	Hold	HK\$7.8	4,237	HK\$0.66	HK\$0.78	16.8x	14.1x	18.0	18.6	1.7x	1.5x
Celestica	CLS US	Buy	US\$112	18,222	US\$5.38	US\$6.64	29.5x	23.9x	25.5	20.4	7.5x	4.8x
Arista	ANET US	Hold	US\$86	133,492	US\$2.75	US\$3.18	38.6x	33.5x	24.9	20.9	9.6x	7.0x
Coherent	COHR US	Hold	US\$64	14,568	US\$3.66	US\$4.39	25.6x	21.3x	15.4	8.5	2.2x	1.8x
Lumemtum	LITE US	Buy	US\$81	6,428	US\$1.86	US\$4.04	49.9x	22.9x	25.3	25.8	12.6x	5.9x
Fabrinet	FN US	Hold	US\$211	10,364	US\$10.08	US\$11.05	28.7x	26.2x	24.4	36.5	7.8x	9.5x
Horizon Robotics	09660 HK	Buy	HK\$9.5	11,388	-CNY 0.01	CNY 0.13	N/A	53.6x	-0.2	2.9	1.8x	1.5x
InnoScience	02577 HK	Buy	HK\$48.0	4,174	-CNY 0.94	-CNY 0.39	N/A	N/A	-24.4	-8.3	11.3x	9.2x

Sources: GF Securities (Hong Kong) Brokerage

Monthly sales and consensus tracking

June monthly data review

TSMC June sales came in at NT\$263.7bn, -18% MoM and +27% YoY, inline with market consensus and our estimate, driven by strong AIGPU/ASIC demand but partially impacted by FX. 2Q revenue of NT\$933.8bn, +11% QoQ and +39% YoY, was also inline with guidance range. Looking ahead, we expect the growth to remain intact, underpinned by sustained AI demand and seasonality. In the long-term, we remain positive towards TSMC, given its technology leadership and better positions in the industry.

AI server accelerating: Hon Hai revenue last month was NT\$540.24bn, -12% MoM, +10% YoY, a miss partly due to unfavorable FX change, while cloud and networking segments was well supported by improving GB200 production. We expect Hon Hai and FII to post good results in 2Q/3Q25, thanks to the ramp up of GB200 with improving margin profile. For consumer products, we are seeing upside to its iPhone build plan and expect more NPI wins, off a low expectation for iPhone 17 cycle. Top 5 ODM reported +5% MoM growth in monthly sales, with a 49% YoY hike. On the other hand, ASIC server momentum is expected to accelerate driven by AWS's Trainium 2. As the primary ODM of Trainium servers, Wiwynn reported another outstanding +21% MoM and +179% YoY on monthly sales, partially thanks to sustained general server momentum into 2Q, while the visibility of general-purpose server for 2H remains mixed for now.

PC uncertainties ahead: NB shipment from top 5 ODMs increased by 2.4mn MoM to 13.0mn, driven by pull-in at the quarter-end and tariff implementation. 2Q PC shipments +8% QoQ and +6% YoY are better than expectations. Given the inventory buildup amid lukewarm markets other than steady commercial refresh, we think the overall 2H demand will be subdued, given sluggish consumer model sales and uninspiring AI PC driven demand. At this stage, PC are still exempt from the reciprocal tariff, while uncertainties regarding tariffs and macro conditions still cloud the outlook. For gaming, GPU demand still remains robust, even with the supply improvement. ASUS and MSI June sales had 8.5%/-5.6% MoM respectively.

Smartphones await seasonality: MediaTek sales in June were NT\$56.4bn, +25% MoM, +31% YoY, off a low base last month amid stable smartphone demand. The company recorded 2Q revenue of NT\$150.4 bn, -1.92% QoQ and +18% YoY, was largely inline with guidance. Meanwhile, Largan reported +28% MoM on monthly sales of NT\$4.2bn, driven by new models buildup with higher ASP. The mgmt. anticipated momentum to further accelerate in the coming months.

Semi lukewarm: For matured nodes, UMC and Vanguard recorded sales YoY of +7.3%/+9%, respectively, while 2Q revenue QoQ of only +1.6%/-1.1% were impacted by FX. In terms of fabless, Realtek's June revenue of NT\$10.2bn was -0.7% MoM and -0.2% YoY, and Novatek's of NT\$8.6bn, down 2.1% MoM and -0.4% YoY. The expectations for electronics are still low lacking of visibility, given customers turned cautious under macro uncertainties.

Figure 3: Taiwan downstream tech monthly sales

TWD m		Jun-25	MoM	YoY	May-25	MoM	YoY	Apr-25	MoM	YoY
EMS										
Hon Hai	2317.TW	540,237	-12.3%	10.1%	615,745	-4.0%	11.9%	641,366	16.3%	25.5%
Pegatron	4938.TW	78,771	-8.1%	11.1%	85,685	-16.7%	-8.0%	102,871	24.7%	14.7%
Quanta	2382.TW	189,868	18.5%	70.6%	160,242	4.0%	58.2%	154,011	2.3%	58.2%
Wistron	3231.TW	209,182	0.4%	135.6%	208,406	55.9%	162.1%	133,704	30.2%	84.1%
Compal	2324.TW	60,520	5.0%	-26.8%	57,625	-7.5%	-25.4%	62,299	-7.8%	-19.4%
Inventec	2356.TW	63,274	3.4%	24.2%	61,203	-1.4%	17.3%	62,100	20.2%	21.6%
PC Brand										
ASUS	2357.TW	68,573	8.5%	17.4%	63,216	12.4%	41.0%	56,218	23.2%	35.1%
Acer	2353.TW	28,685	49.7%	1.7%	19,168	3.1%	-9.3%	18,595	8.9%	4.2%
Graphics Card										
MSI	2377.TW	19,561	-5.6%	33.0%	20,715	-1.6%	33.2%	21,041	26.6%	28.3%
Gigabyte	2376.TW	25,448	-45.5%	11.0%	46,728	55.3%	107.9%	30,098	34.8%	5.5%
PCB/FPC										
Zhen Ding	4958.TW	12,823	8.8%	19.7%	11,791	-13.2%	10.8%	13,589	11.3%	22.9%
Unimicron	3037.TW	10,929	2.9%	23.2%	10,621	-2.7%	10.1%	10,916	9.3%	16.6%
Compeq	2313.TW	5,717	-4.4%	7.4%	5,983	-8.0%	4.9%	6,501	25.0%	11.0%
Flexium	6269.TW	1,733	-3.2%	-14.6%	1,791	-9.2%	-32.5%	1,973	39.6%	-31.6%
Optics										
Largan	3008.TW	4,146	28.0%	2.6%	3,238	-26.0%	-7.1%	4,375	-7.6%	26.5%
Genius	3406.TW	1,664	33.1%	7.0%	1,251	-32.6%	-19.6%	1,854	33.4%	-1.2%
Acoustics										
Merry	2439.TW	3,455	-2.1%	5.8%	3,528	7.2%	0.4%	3,292	20.9%	-6.3%
Primax	4915.TW	4,930	1.8%	-5.7%	4,841	0.4%	-5.3%	4,822	-5.8%	-4.1%
Components										
Catcher	2474.TW	1,791	0.9%	14.3%	1,775	18.2%	8.4%	1,502	8.5%	2.3%
SZS	3376.TW	968	0.5%	-15.3%	964	-4.4%	-18.6%	1,008	16.9%	-12.6%
Server										
Wiwynn	6669.TW	85,808	20.6%	179.0%	71,133	11.5%	187.3%	63,803	37.2%	190.5%
Aspeed	5274.TW	782	3.0%	55.0%	760	7.9%	74.5%	704	15.6%	69.4%
Chenbro	8210.TW	1,635	-10.8%	23.3%	1,833	-7.2%	46.9%	1,975	51.5%	91.0%
KingSlide	2059.TW	1,386	-0.5%	62.4%	1,394	-3.8%	60.5%	1,449	8.2%	82.3%
Networking										
Sercomm	5388.TW	4,257	11.6%	-4.9%	3,814	-8.0%	-11.1%	4,145	10.9%	-16.2%
Wistron NeWeb	6285.TW	8,607	-4.6%	-17.1%	9,025	-8.5%	-12.7%	9,859	12.3%	-1.4%

Sources: TWSE

Figure 4: Taiwan upstream tech monthly sales

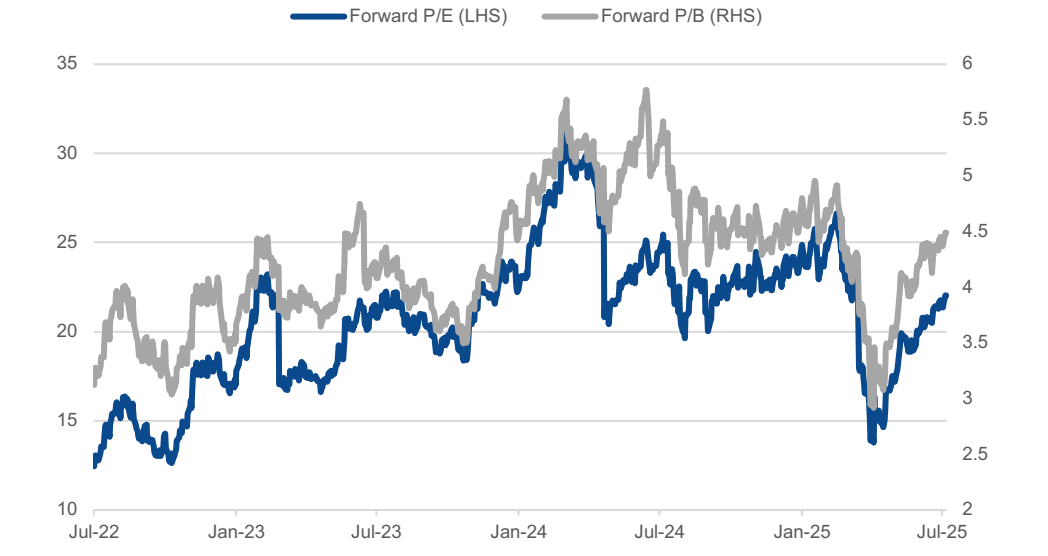
TWD m		Jun-25	MoM	YoY	May-25	MoM	YoY	Apr-25	MoM	YoY
Foundry										
TSMC	2330.TW	263,709	-17.7%	26.9%	320,516	-8.3%	39.6%	349,567	34.4%	48.1%
UMC	2303.TW	18,823	-3.4%	7.3%	19,480	-4.8%	-0.2%	20,455	12.4%	3.6%
Vanguard	5347.TW	4,474	26.0%	9.0%	3,550	-6.4%	-0.6%	3,793	-4.1%	11.9%
OSAT										
ASE	3711.TW	49,513	1.0%	5.5%	49,027	-6.1%	3.2%	52,211	16.1%	13.9%
Powertech	6239.TW	6,263	3.0%	-4.1%	6,080	6.3%	-7.3%	5,717	14.9%	-12.0%
Chipbond	6147.TW	1,801	-0.6%	0.7%	1,812	2.9%	3.3%	1,760	-3.6%	9.3%
Chipmos	8150.TW	1,850	-8.7%	-3.5%	2,025	8.8%	0.1%	1,861	-8.4%	-0.4%
IC Design										
MediaTek	2454.TW	56,434	24.9%	31.0%	45,181	-7.3%	7.2%	48,754	5.6%	16.0%
Novatek	3034.TW	8,425	-2.1%	-0.4%	8,607	-5.6%	2.0%	9,119	-1.6%	9.5%
Realtek	2379.TW	10,208	-0.7%	-0.2%	10,276	-10.1%	-1.0%	11,430	1.0%	13.5%
Aspeed	5274.TW	782	3.0%	55.0%	760	7.9%	74.5%	704	15.6%	69.4%
Silergy	6415.TW	1,512	4.1%	-4.2%	1,452	-1.9%	-6.1%	1,481	11.0%	-1.2%
GaAs										
WIN SEMI	3105.TW	1,364	9.0%	-18.1%	1,251	7.4%	-24.3%	1,165	-0.5%	-29.1%
AWSC	8086.TW	351	8.7%	-17.6%	323	7.8%	-25.6%	300	15.2%	-32.8%
VPEC	2455.TW	264	20.5%	-11.1%	219	-5.3%	-25.0%	231	-12.9%	-19.3%
MLCC										
Yageo	2327.TW	11,056	8.3%	10.5%	10,211	-11.2%	-4.7%	11,503	17.9%	7.5%
Walsin	2492.TW	3,148	-0.9%	10.4%	3,176	-0.5%	5.6%	3,193	15.0%	6.2%
Display										
AUO	2409.TW	21,922	-9.3%	-13.1%	24,177	4.5%	-4.6%	23,138	-6.0%	-2.6%
Innolux	3481.TW	18,511	-1.1%	-1.2%	18,724	-1.4%	-0.7%	18,997	2.7%	-1.3%
Memory										
Nanya	2408.TW	4,074	22.2%	21.1%	3,333	6.9%	-0.5%	3,119	37.2%	-2.7%
Winbond	2344.TW	7,105	0.2%	-3.7%	7,094	4.0%	-0.4%	6,819	7.3%	-2.3%
Macronix	2337.TW	2,201	-3.0%	1.1%	2,268	-2.7%	5.0%	2,330	23.0%	10.0%

Sources: TWSE

Tech valuation in major markets

Philadelphia Semiconductor Index (SOX) forward P/E and P/B increased from 20.4x/4.3x last month to 22x/4.5x, respectively, continuing to benefit from broad market rally.

Figure 5: SOX Index's Forward P/E, Forward P/B



Sources: Bloomberg

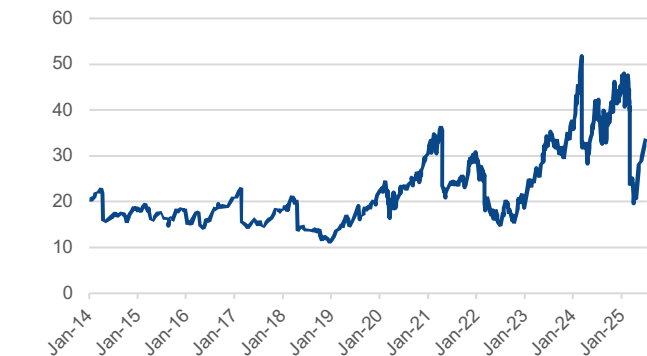
In the meantime, global semi forward P/E increased from 30.5x to 34.2x now, while global hardware forward P/E edged higher from 25.7x to 26.7x.

Figure 6: MSCI World Tech Hardware Forward P/E



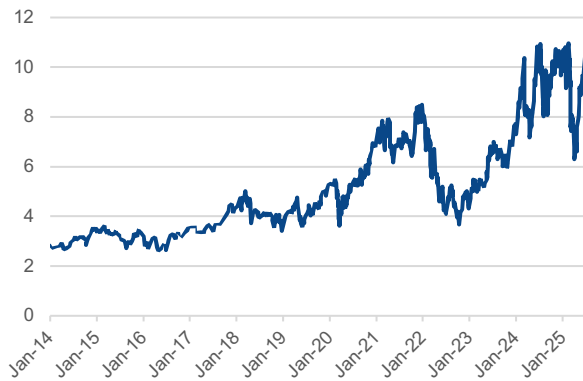
Sources: Bloomberg

Figure 7: MSCI World Semi Forward P/E



Sources: Bloomberg

In terms of P/B, the global semi forward P/B rose to 10.8x from 9.6x last month, while China semiconductor P/B remained largely unchanged at 4.8x.

Figure 8: MSCI World Semi Forward P/B


Sources: Bloomberg

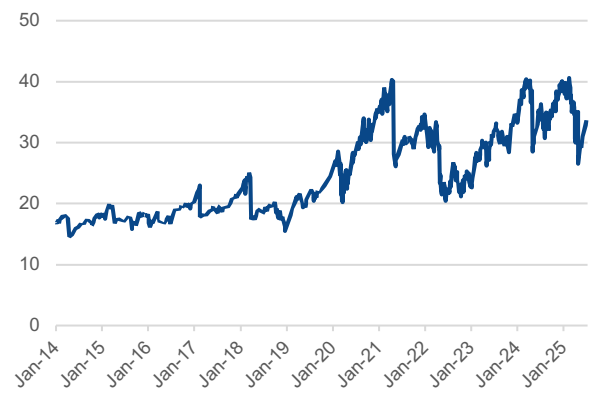
Figure 9: SWS Semiconductor Index P/B


Sources: Wind

China IT sector P/E climbed to 29.7x from 27.8x, which is at the upper end of historical average of 20-30x. In terms of US IT sector, the one year forward P/E also drifted up to 33.8x from 31.3x.

Figure 10: MSCI China IT Forward P/E


Sources: Bloomberg

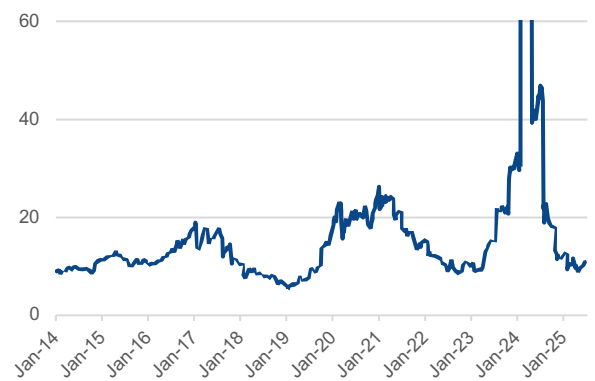
Figure 11: MSCI US IT Forward P/E


Sources: Bloomberg

The Taiwan IT sector P/E rose to 18.0x from 16.5x, above the historical average of 16x, and Korean IT sector P/E expanded to 9.8x from 8.8x.

Figure 12: MSCI Taiwan IT Forward P/E


Sources: Bloomberg

Figure 13: MSCI Korea IT P/E


Sources: Bloomberg

Tariff Implications to Tech

Rising Tariffs

Back in April and May, Trump declared that the 10-125% reciprocal tariffs would be deferred to July 9, casting uncertainty over the preceding three months. On July 2, a week before the scheduled date, Trump reduced the reciprocal tariff rate to 20% for Vietnamese products and to 40% for goods transiting through Vietnam according to the bilateral agreement, with these changes set to take effect on August 1. Since then, the U.S. government has been actively revising tariff rates and pushing for negotiations with various countries before July 9, yet with limited progress. The market has reacted calmly to these moves, with S&P 500 index maintaining a mild upward trend, given the moderated tariff rates compared to initial proposals have eased some concerns.

On July 7, Trump announced a second postponement of reciprocal tariffs until August 1, while setting updated reciprocal tariff rate of 25-40% for 14 countries, with Japan and South Korea each facing a 25% rate. Additionally, on July 8, Trump escalated trade tensions by threatening new tariffs on semiconductors and countries aligning with the BRICS. Subsequently, on July 9 and 11, he imposed a 50% tariff on Brazil and a 35% tariff on Canada, both effective August 1. However, market reaction remained muted again, with daily fluctuation of less than 1%. Then on July 12, Trump announced a 30% tariff on goods imported from the European Union (EU) and Mexico, effective August 1, 2025.

On the technology front, though semiconductor products are still under Section 232 investigation, consumer electronics continue to be exempted from reciprocal tariffs at this stage. Near term, the market will watch the tariff rate which may rise if threatened tariffs go into effect August 1st as scheduled. Moreover, a 50% tariff on copper and yet-to-be-announced tariff on semiconductors will also raise the effective tariff. Semiconductor tariffs are yet to be announced and it is unclear if the administration will also seek to tariff a broader array of electronics products.

Figure 14: S&P 500 Index (normalized) comparison: 2018 vs 2025



Sources: Bloomberg

Figure 15: Timeline of U.S. Tech-Related Tariffs

Date	Countries	Applicable Products	Details	Status
Feb 4	China	All	10% tariffs under IEEPA	In effect
Mar 4	Canada & Mexico	All	25% tariffs under IEEPA	In effect
Mar 4	China	All	10% more tariffs under IEEPA, 20% in total	In effect
Apr 1		Semiconductors & Manufacturing Equipment	Section 232 investigation	Underway
Apr 3	All	Automobiles	25% Section 232 Tariffs	In effect
Apr 5	All	All	10% Reciprocal Tariffs	Till Apr 10
Apr 9	All	All	10-84% Elevated Reciprocal Tariffs, 104% in total for China	Paused for 90 days except for China
Apr 10	China	All	125% Elevated Reciprocal Tariffs, 145% in total for China	Till May 14
May 3	All	Auto parts	25% Section 232 Tariffs	In effect
May 8	UK	Automobiles	10% Tariffs (reduced from 25%)	In effect since Jun 30
May 14	China	All	10% Reciprocal Tariffs	10% in effect, others paused for 90 days
May 23	EU	All	50% Reciprocal Tariffs	Delayed to Jul 9
May 31	China	Autos, batteries, critical minerals, solar cells, semiconductors	Exclusions for Section 301 Tariffs extended to Aug 31	Extended to Aug 31
Jul 2	Vietnam	All	20% Tariffs for export 40% Tariffs for trans-shipment	Start from Aug 1
Jul 7	All	All	Reciprocal Tariffs exemption period till Jul 9	Extended to Aug 1
Jul 7	14 countries	All	Tariff rate revision: 25% (Japan & Korea) - 40%	Start from Aug 1
Jul 11	Canada	All	35% Reciprocal Tariffs	Start from Aug 1
Jul 12	EU & Mexico	All	30% Reciprocal Tariffs	Start from Aug 1

Sources: Bloomberg

Downstream tech – Smartphone & Apple

Maintain global smartphone units at +1%/+2% in 2025/2026

According to IDC's forecast on June 4, it lowered the global smartphone shipment forecast from 2.3% YoY to 0.6% YoY in 2025 to 1.24bn units. Meanwhile, the five-year compound annual growth rate (CAGR) for global smartphone shipments is forecasted to be 1.4% through 2029. IDC said this meager growth is due to increasing smartphone penetration rates, lengthening refresh cycles and cannibalization from used smartphones. On the other hand, IDC expects a 4% growth in average selling prices as new premium devices are launched in the second half of the year. China, meanwhile, is forecast to grow 3% YoY driven by government subsidies that will stimulate demand and boost Android devices, mostly. On the other hand, we expect a pull-in from Apple in production ahead of the potential tariff policies.

Regarding the Edge AI, there are not yet any clear killer applications that would stimulate replacement demand. Our estimates therefore do not incorporate expectations of excess replacement demand in 2025, and we instead assume a gradual increase in global demand for smartphones.

Figure 16: Global Smartphone shipment forecast

Shipments (mn)	2Q23	3Q23	4Q23	1Q24	2Q24	3Q24	4Q24	1Q25	2024	YoY	2025E	YoY
Global smartphone	268	304	324	300	292	316	332	305	1240		1255	
YoY	-6.3%	0.6%	6.8%	11.9%	8.9%	4.0%	2.3%	1.5%	6.5%		1.2%	
By OEMs												
Samsung	53.5	59.5	53.0	60.1	53.8	57.8	51.7	60.6	223	-1%	218	-2%
Apple	44.5	54.1	80.2	52.6	45.7	56.0	76.9	57.9	231	-1%	232	0%
Huawei	9.6	8.1	10.6	12.5	13.7	11.2	11.0	11.0	48	39%	50	3%
New Honor	13.8	17.7	17.1	16.8	15.3	15.0	17.0	17.0	64	4%	65	2%
OPPO/Realme	35.5	37.8	35.4	35.3	38.1	42.8	39.1	35.5	155	8%	165	6%
Vivo	21.3	22.0	24.0	21.3	25.9	27.0	26.0	22.7	100	14%	105	5%
Xiaomi	33.2	41.5	40.7	40.8	42.3	42.8	42.7	41.8	169	15%	178	6%
Share %												
Samsung	20%	20%	16%	20%	18%	18%	16%	20%	18%		17%	
Apple	17%	18%	25%	18%	16%	18%	23%	19%	19%		18%	
Huawei	4%	3%	3%	4%	5%	4%	3%	4%	4%		4%	
New Honor	5%	6%	5%	6%	5%	5%	5%	6%	5%		5%	
OPPO/Realme	13%	12%	11%	12%	13%	14%	12%	12%	13%		13%	
Vivo	8%	7%	7%	7%	9%	9%	8%	7%	8%		8%	
Xiaomi	12%	14%	13%	14%	14%	14%	13%	14%	14%		14%	

Sources: IDC, GF Securities (Hong Kong) Brokerage

Raise iPhone 17 build plan due to the production acceleration on tariff

According to IDC, Apple had the best Q1 ever in terms of units shipped, stockpiling to avoid the U.S. tariffs, but also to other regions, as the channels fear that supply chain disruption could lead to a shortage of stock and price increases. Its performance in China declined, as Apple's Pro models were exempt from the Chinese government subsidy program.

However, the launched budget iPhone 16e appears to be a mute product cycle due to inferior price to performance, despite the fact that it is qualified for China subsidy program. Hence, we expect an inline 2Q25 iPhone shipments.

For 2H25, we raised our iPhone 17 build plan from 81m to 86m, as we believe that Apple may accelerate the production amid tariff uncertainties.

iPhone 17 spec forecast update

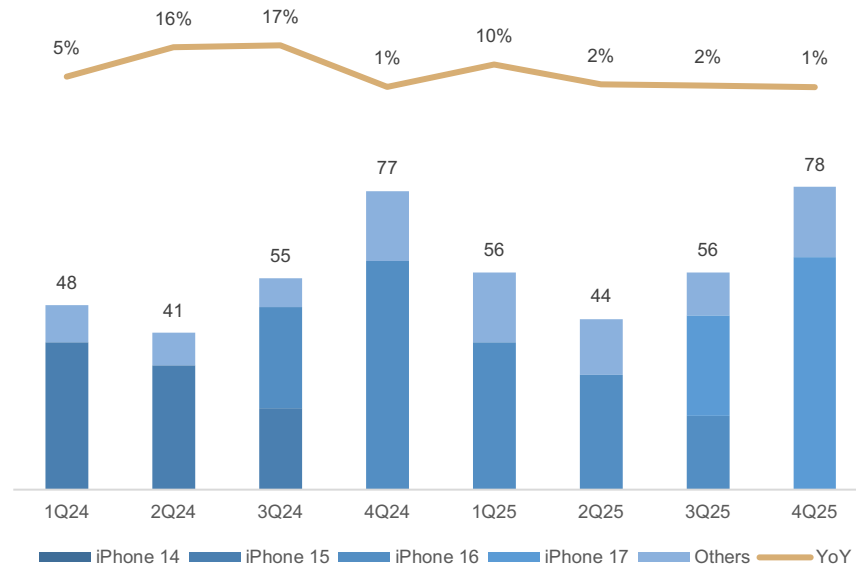
For iPhone 17, we keep our previous update, highlighted in the below figure, with key specs of aluminum frame (except the iPhone 17 Air adopting titanium), A19 Pro chip for iPhone 17 Pro and Pro Max, Apple-designed Wi-Fi 7, Apple-designed baseband C1 for the iPhone 17 Air, 24MP 6P front camera, 48MP rear Telephoto camera for the iPhone 17 Pro models, 12GB DRAM LPDDR5X for the iPhone 17 Pro and Pro Max, 12GB LPDDR5 for iPhone 17 Air, and 8GB LPDDR5 for iPhone 17. Yet, we now expect the iPhone 17 model to be equipped with A19 (vs. earlier expectation of A18). Overall, the spec upgrade appears lukewarm.

Figure 17: iPhone 17 series spec expectations

	iPhone 17	iPhone 17 Air	iPhone 17 Pro	iPhone 17 Pro Max
Launch	Sep-25	Sep-25	Sep-25	Sep-25
Display	6.1"	6.6"	6.3"	6.9"
Processor	A19 N3P	A19 N3P	A19 Pro, N3P	A19 Pro, N3P
DRAM	LPDD5 8GB	LPDD5 12GB	LPDD5X 12GB	LPDD5X 12GB
Front camera	24MP, 6P	24MP, 6P	24MP, 6P	24MP, 6P
Rear camera	48MP 7P, 12MP 5P	48MP 7P	48MP 7P, Periscope 48MP 1G+3P, 48MP 6P	48MP 7P, Periscope 48MP 1G+3P, 48MP 6P
Face ID	Structured light	Structured light	Structured light	Structured light
Casing	Aluminum	Titanium	Aluminum	Aluminum
BB	Qualcomm	Qualcomm or Apple	Qualcomm	Qualcomm
Charging	35W	35W	35W	35W
NPI	Foxconn	Foxconn	Foxconn	ICT

Sources: MacRumors, GF Securities (Hong Kong) Brokerage

Figure 18: Apple iPhone build plan and YoY (mn units)



Sources: Company data, IDC, GF Securities (Hong Kong) Brokerage

Downstream tech – PC

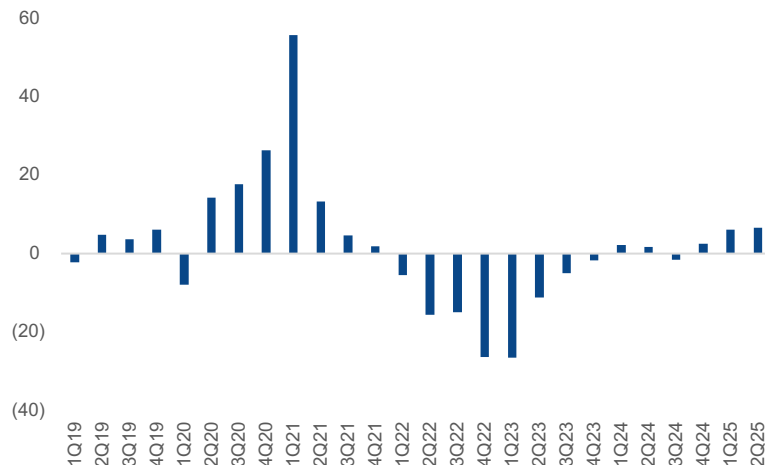
Inventory buildups to weigh on 2H outlook

On Jul 8, IDC reported 2Q25 PC shipment of 68.4mn, +6.5% YoY, mainly attributed to pull-ins ahead of tariff implementation, with momentum extending to 2Q due to temporary exemption from reciprocal tariff. Among vendors, Lenovo further widened its lead with a 24.8% share, up 0.7ppts from a quarter ago, while ASUS also gained 0.9ppts share, well supported by the new gaming models. In contrast, Dell's share continued to be eroded, with only 14.3% in the quarter.

Segment-wise, we have seen solid gaming models sales, fueled by the launch of new NVDA's RTX 50 and AMD Radeon GPUs, along with robust demand in China amid large gaming user base and new AAA games releases. On the other hand, commercial refresh remains on track driven by Win 10 EOL, with OEMs anticipating sustained momentum of replacement cycle into 2026. Nevertheless, despite the solid shipment in 2Q, underlying consumer demand remains tepid, particularly for AI PCs, which are experiencing a slower-than-expected penetration.

In response to tariffs, vendors such as HP and Lenovo have raised notebook prices in the U.S. to offset increased costs. However, given high inventory levels, macro uncertainties, and fluid tariff policies, we expect the growth for 3Q/4Q25 to be constrained, coming off a stronger than expected 1H. We think there would be more promotions to digest the inventory in 2H. As such, we forecast only a +2% YoY shipment growth in 2025 vs. IDC's forecast of +4%, with limited visibility into 2026.

Figure 19: Global PC shipment YoY (%)

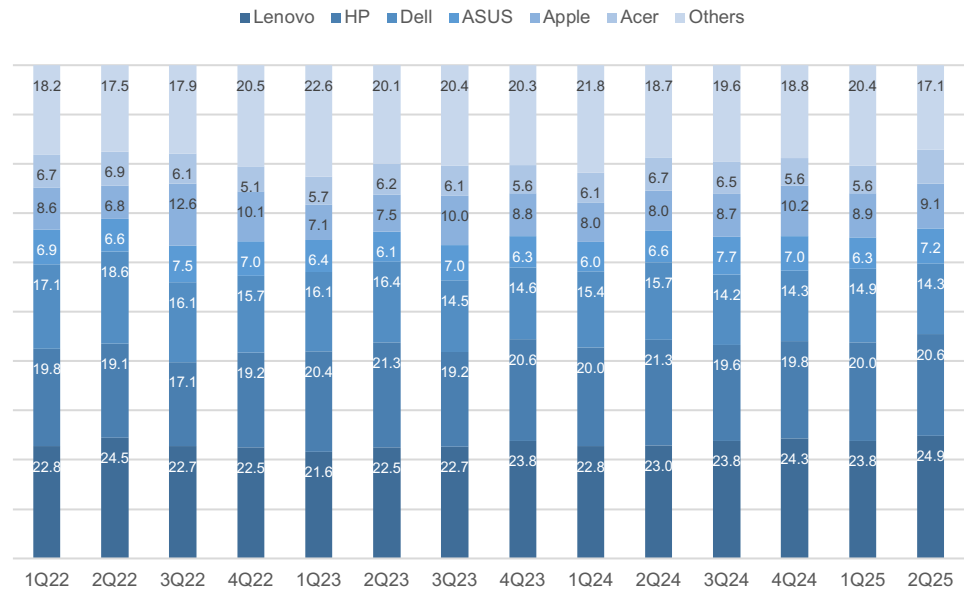


Sources: IDC, GF Securities (Hong Kong) Brokerage

Figure 20: Global PC shipment (mn units)



Sources: IDC, GF Securities (Hong Kong) Brokerage

Figure 21: Global PC market share (%) by brands


Sources: IDC, GF Securities (Hong Kong) Brokerage

Figure 22: Monthly Notebook PC ODM shipment

Unit mn	Jun-24	Jul-24	Aug-24	Sep-24	Oct-24	Nov-24	Dec-24	Jan-25	Feb-25	Mar-25	Apr-25	May-25	Jun-25
Quanta	4.5	3.4	4.3	4.9	3.2	3.6	4.3	2.8	3.1	4.9	3.3	3.8	5.0
Compal	3.0	2.5	2.9	2.9	2.8	2.7	2.3	2.0	2.1	2.9	2.2	2.4	2.5
Wistron	1.8	1.6	1.7	1.9	1.8	1.7	1.6	1.5	2.2	1.9	1.7	1.8	2.4
Pegatron	0.8	0.8	0.9	0.8	0.6	0.6	0.8	0.6	0.8	0.7	0.6	0.8	1.0
Inventec	1.7	1.7	1.7	1.8	1.7	1.7	2.0	1.5	1.6	1.9	1.7	1.8	2.1
Total	11.8	10.0	11.5	12.3	10.1	10.3	11.0	8.4	9.8	12.3	9.5	10.6	13.0
YoY	-7%	-8%	-2%	-1%	6%	-4%	10%	0%	18%	3%	-4%	-4%	10%
MoM	7%	-15%	15%	7%	-18%	2%	7%	-24%	16%	26%	-23%	11%	23%

Source: Company data

Upstream tech – AI Semi & Supply Chain

2025 CSP CAPEX Intact; Remaining positive on 2026 outlook

During the CY1Q25 earnings, the top 4 CSPs delivered better-than-expected CY1Q25 results, with cloud revenues generally inline supported by rising AI demand, which helped reverse the negative market sentiment. In addition, their quarterly CAPEX met with expectations, with 2025 full-year CAPEX guidance unchanged except for Meta, which raised its guidance from to a higher range of \$64–\$72bn.

For 2026, given MSFT's strategy to build a unified agent platform with Copilot integration, along with sustained AI ARR, we believe its mid-term CAPEX plan would be structural and sustainable. As such, we expect **MSFT's FY26 CAPEX to grow by >10% YoY, exceeding US\$90 bn**. Overall, given solid cloud momentum, accelerating AI contribution, and positive commentary on future investments, we **reiterated our forecast of ~15% YoY CAPEX growth for the top 4 CSP in 2026**.

Figure 23: Major Cloud players Capex

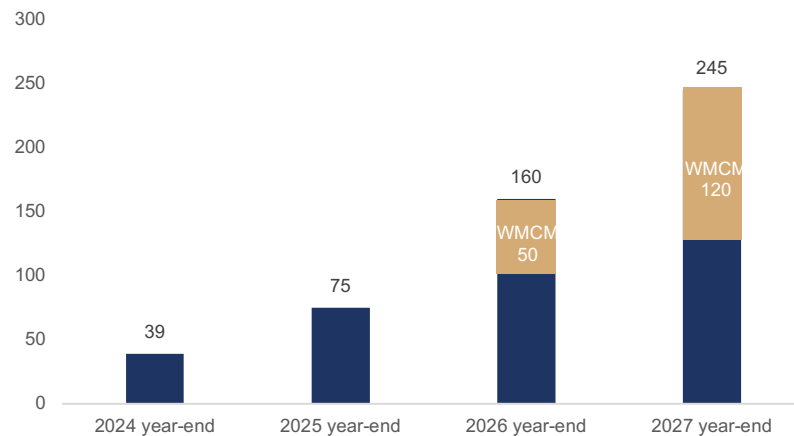
Capex (USD m)	2022	2023	2024	2025E
Microsoft	28,400	41,200	75,600	92,181
Amazon	63,645	52,729	82,999	106,058
Alphabet	31,485	32,251	52,535	77,398
Meta	31,900	28,200	39,200	68,000
Total	155,430	154,380	250,334	343,637
YoY	2022	2023	2024	2025E
Microsoft	3%	45%	83%	22%
Amazon	4%	-17%	57%	28%
Alphabet	26%	2%	63%	47%
Meta	67%	-12%	39%	73%
Total	17%	-1%	62%	37%

Sources: Bloomberg, GF Securities (Hong Kong) Brokerage

TSMC's CoWoS capacity to remain aggressive

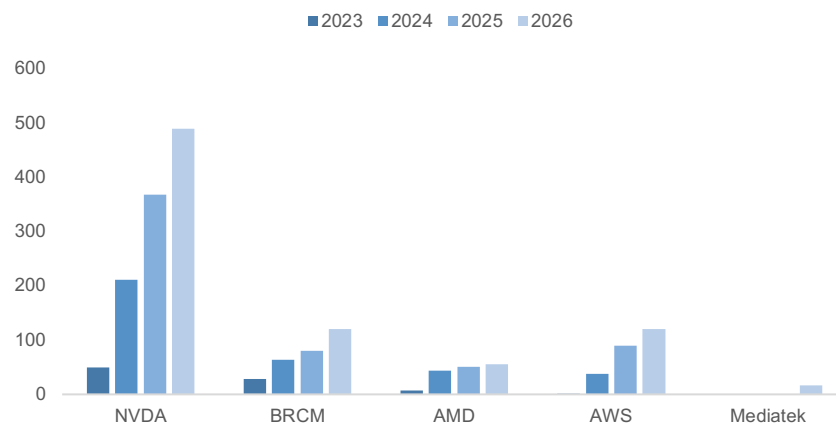
As stated in our previous report, Apple's iPhone 18 Fold, 18 Pro Max and 18 Pro will likely debut the A20 chip utilizing TSMC's Wafer-Level Multi-Chip Module (WMCM) packaging alongside N2 process technology. WMCM integrates multiple dies—such as logic SoC and DRAM—directly on wafer before dicing and adopts RDL to replace interposer for better signal integrity and thermal performance. TSMC will establish a dedicated WMCM production line at its AP7, leveraging equipment and process similar to CoWoS-L without on-substrate. We see TSMC is preparing capacity of up to 50KPM by end of 2026 and estimate the capacity to reach 110-120KPM by end-2027.

Meanwhile, we expect a clear ASIC demand in 2026. We forecast TSMC's CoWoS-L+S capacity to be 110-115KPM by end-2026, from 75KPM in end-2025.

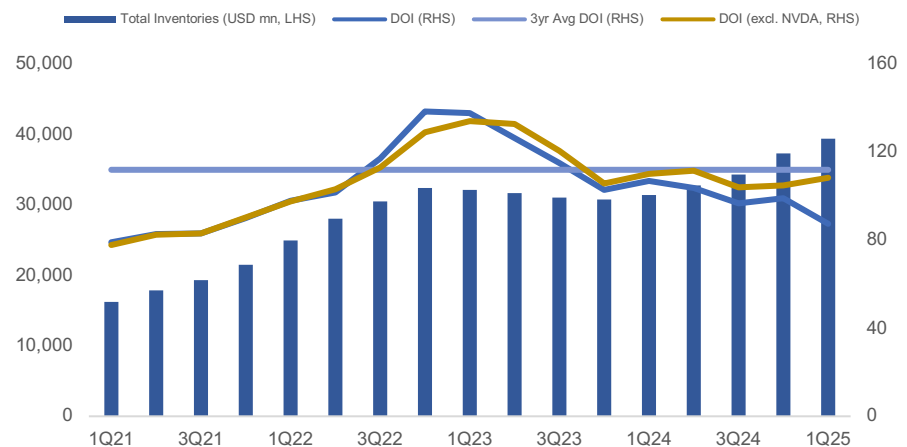
Figure 24: TSMC's quarterly CoWoS + WMCM capacity (kwpm) trend


Sources: Company data, GF Securities (Hong Kong) Brokerage

We maintain our CoWoS build plan for most AI fabless companies while raised AWS CoWoS from 85k to 90k for 2025, driven by R2.

Figure 25: Major fabless CoWoS build plan (k wafers)


Sources: Company data, GF Securities (Hong Kong) Brokerage

Figure 26: Top-15 fabless DOI


*Including NVDA, AMD, AVGO, QCOM, MTK, MRVL, QRVO, Novatek, Realtek, SWKS, HIMX, WILL, MPWR, IFX, NXP
Sources: Company data, GF Securities (Hong Kong) Brokerage

Upstream tech – Memory & HBM

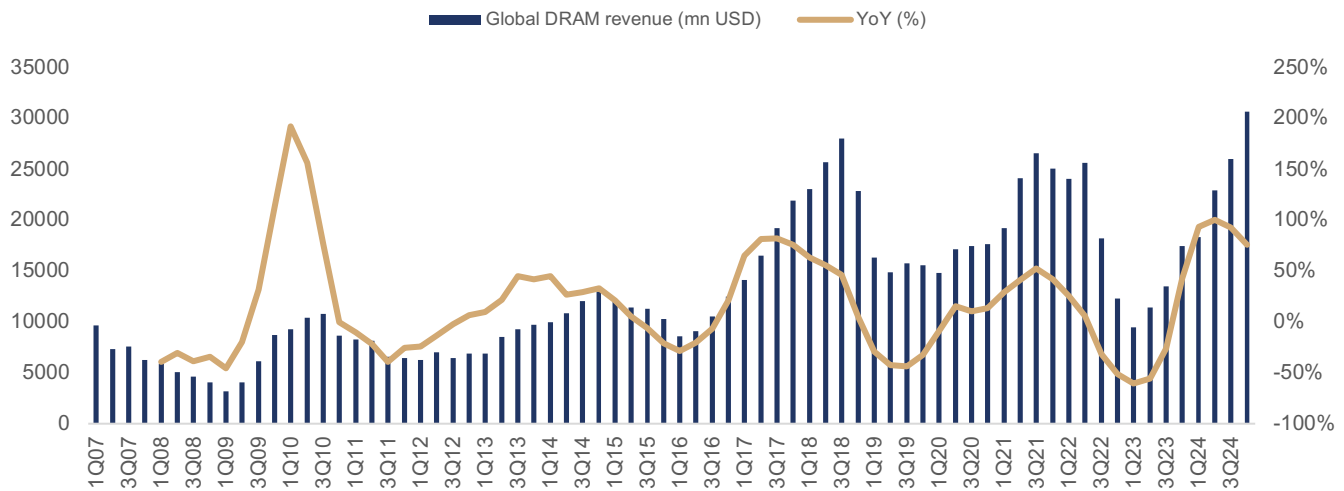
Better pricing in 3Q25; DDR4 cycle likely ending in 4Q25

In our Micron preview, we initially forecasted a 7% QoQ increase in blended DRAM contract prices (ex-HBM) for 3Q25, above consensus estimates of ~5%, but the actual price hike could now exceed 10% due to stronger-than-expected DDR4 pricing driven by Micron, Samsung, and SK Hynix's EOL plans. PC DDR4 prices are up ~40% on tariff-related restocking, while smartphone/server DDR4 price up 28~33%, with LPDDR4 at the lower end due to CXMT's smartphone-centric supply. However, our checks suggest PC restocking will end in 4Q25, flattening PC DDR4 prices, while LPDDR4 and server DDR4 pricing will catch up but moderate overall. We expect DDR4 price hikes to end in 4Q25, as further increases could incentivize major memory makers to switch back to DDR4 production.

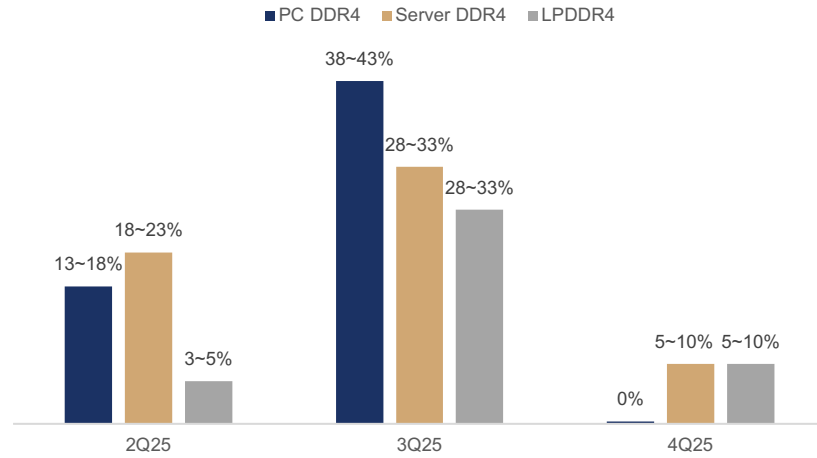
2026 DDR5 pressured on local supply

CXMT is likely to phase out DDR4 production and reallocate capacity to DDR5. We expect DDR5 bit shipments to reach 30% of its total output by year-end, with LPDDR5X supporting 16M units 12GB smartphones (assuming 50% yield) and server-grade products already in volume production. By 2025, CXMT's capacity will reach 280KPM and we expect the local memory maker to represent 13% of global DRAM capacity even excluding the ~50-60KPM reserved for Gigaevic's specialty DRAM outsourcing. That said, we expect CXMT to represent 8~9% of DDR5 bit shipment in 2026, up from 2~3% in 2025, and significantly pressure DDR5 pricing.

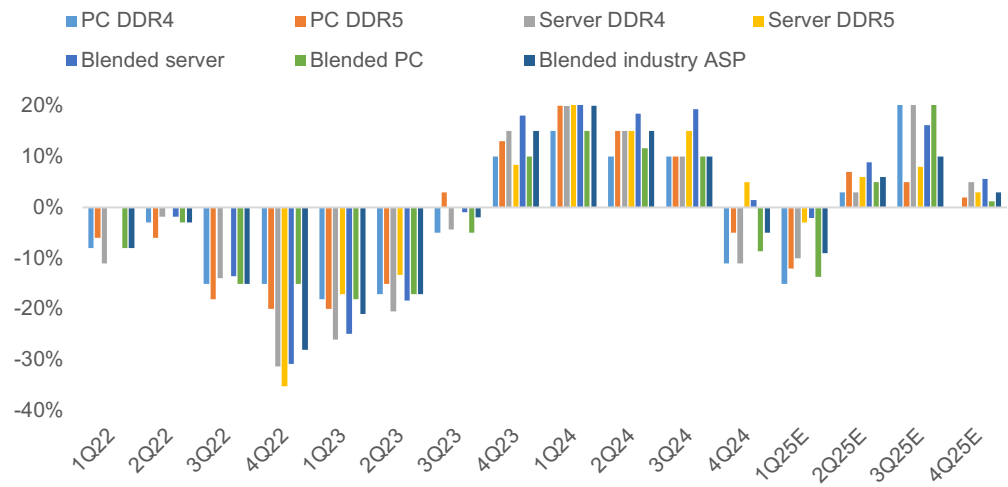
Figure 27: Global DRAM revenue and YoY (USD mn)



Source: TrendForce, DRAMeXchange, GF Securities (Hong Kong) Brokerage

Figure 28: DDR4 price forecast


Source: TrendForce, GF Securities (Hong Kong) Brokerage

Figure 29: DRAM price forecast


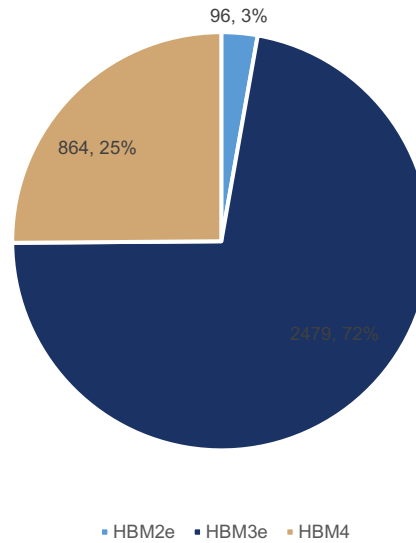
Source: TrendForce, GF Securities (Hong Kong) Brokerage

Figure 30: HBM demand forecast

	Shipment (in K units)			HBM type	HBM content (GB)	HBM demand in (K GB)		
	2024	2025	2026			2024	2025	2026
Nvidia								
A100				HBM2e	80	0	0	0
H20	1500	900		HBM3	96	144000	86400	0
H100	1200			HBM3	80	96000	0	0
H200	2000	300		HBM3e	141	282000	42300	0
B200	200	2500		HBM3e	192	38400	480000	0
B Ultra		3200	5500	HBM3e	288	0	921600	1584000
R series		0	2000	HBM4	288	0	0	576000
Total	4900	6900	7500			560400	1530300	2160000
AMD								
Mi300	530	300		HBM3	192	101760	57600	
M325		100		HBM3e	288	0	28800	
M350		350	400	HBM3e	288	0	100800	115200
M400			400	HBM3e	288			172800
Total	530	750				101760	187200	288000
Google								
TPU v5e	1600	600		HBM2e	16	25600	9600	0
TPU v5p	1200	800		HBM2e	96	115200	76800	0
TPU v6	0	1200	2000	HBM3	32	0	38400	64000
TPU v7		425	595	HBM3e	192	0	81600	114240
TPU v8p			340	HBM3e	288	0	0	97920
TPU v8e			330	HBM3e	216	0	0	71280
Total	2800	3025	2000			140800	206400	347440
Amazon								
Inferentia 2	700			HBM2e	32	22400	0	0
Inferentia 2.5	50	200		HBM3	48	2400	9600	0
Trainium 2	300	1400		HBM3	96	28800	134400	0
R2			1000	HBM3e	96		0	96000
Trainium 3			1300	HBM3e	128			166400
Total	1050	1600	2300			53600	144000	262400
Huawei								
910B	500	480		HBM2e	64	32000	30720	
910C	10	320	1000	HBM2e	96	960	30720	96000
Total	510	800	1000		160	32960	61440	96000
Other GPU								
Gaudi	100	150	200	HBM3	96	9600	14400	19200
Total	100	150	200		96	9600	14400	19200
MSFT								
M300			400	HBM4	288			115200
Total			400		288			115200
Meta								
MTIA2		500		NA	NA			
MTIA3			700	HBM3e	216			151200
Total		500	700		216			151200
Total HBM demand in M GB						890	2144	3439

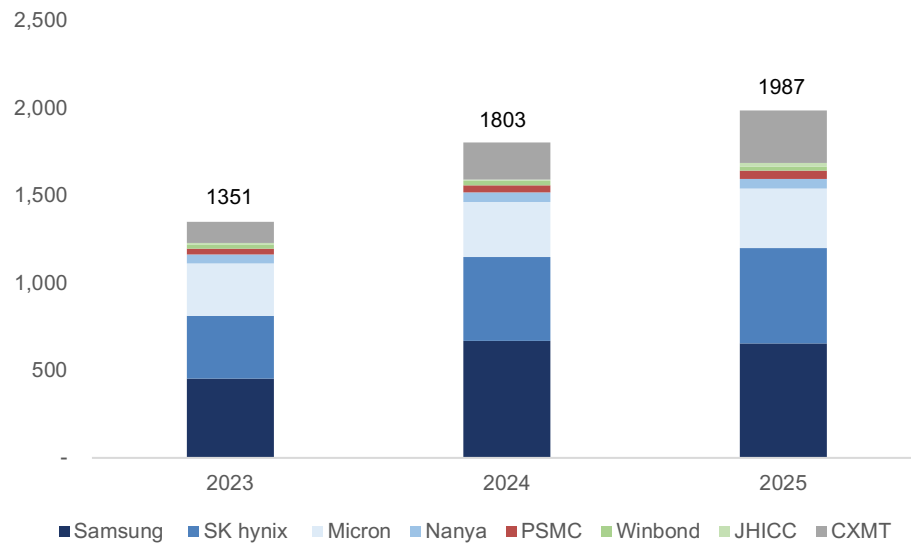
Source: Bloomberg, GF Securities (Hong Kong) Brokerage

Figure 31: 2026 HBM demand by product type (mn GB, % of total HBM demand)



Source: Nature Nanotechnology, GF Securities (Hong Kong) Brokerage

Figure 32: 2023-2025 year-end memory capacity (kwpm)



Sources: Company data, GF Securities (Hong Kong) Brokerage

Upstream tech – WFE

ASML's booking likely another disappointment

We forecast ASML's 2Q25 order intake at ~€5bn (including ~€2.3bn of EUV booking), below Asian view of ~€6.5-7bn. Though market is turning more positive on TSMC's order on accelerating Arizona fab expansion, we see TSMC's 2026 EUV demand remain in 14-16 units range. Conversely, Intel remains absent from 2Q25 bookings, with its 2026 EUV demand likely to decline sharply to 3-5 units compared to 10 units in 2025. On the DUV front, orders from China remain robust at ~€2 billion, driven primarily by Huali's capacity expansion.

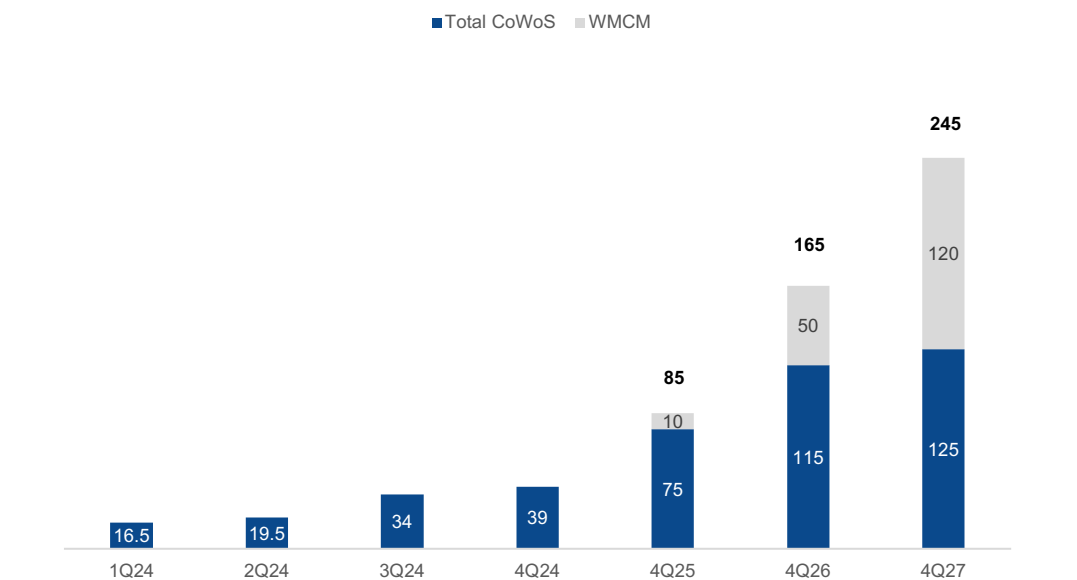
We maintain a cautious outlook WFE with our growth forecast at 5%/flat for 2025/2026. Our 2026 EUV demand estimate remains unchanged at 44 units. While 2027 appears more promising, driven by tool move-in at Samsung's Taylor fab, TSMC's Arizona expansion, and Intel's 18A ramp, we believe the market will need to adjust 2026 earnings expectations downward. Current consensus still implies 5% YoY revenue growth for major SPES.

Stronger backend demand on CoWoS and WMCM

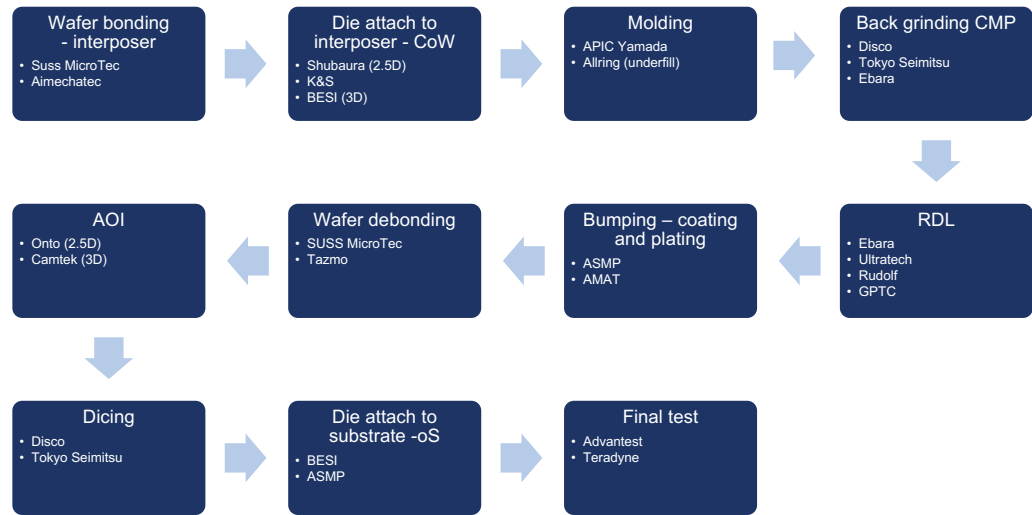
We believe CoWoS and WMCM equipment providers will outperform on stronger 2H26 capacity build. We believe WMCM is a new and strong demand driver, as it suggests the CoW technology being expanded to smartphone/tablets.

- **Grand Process (3131 TT)** is a key beneficiary, given its dominance in wet cleaning tools for batch, single-wafer, and post-CMP processes. We believe share price driver is backlog, and consensus expects 150/150 machines in 2025/2026 (with likely upside from WMCM of ~40 machines in 2026). The emerging trends of WMCM, CPO and CoPoS will be drivers in 2027.
- **Onto Innovation (ONTO US)** remains appealing for 2.5D inspection despite some share losses. Onto is also expanding its product portfolio into 3D inspection and litho for CoPoS.
- **Shibaura (6950 JP)**'s bonder demand will be more than doubled in WMCM for VBA (micro bumping), making Shibaura the best play in WMCM. However, we expect order to be more reflected in 2026 as the mgmt. guided order to be yoy down in 2025.

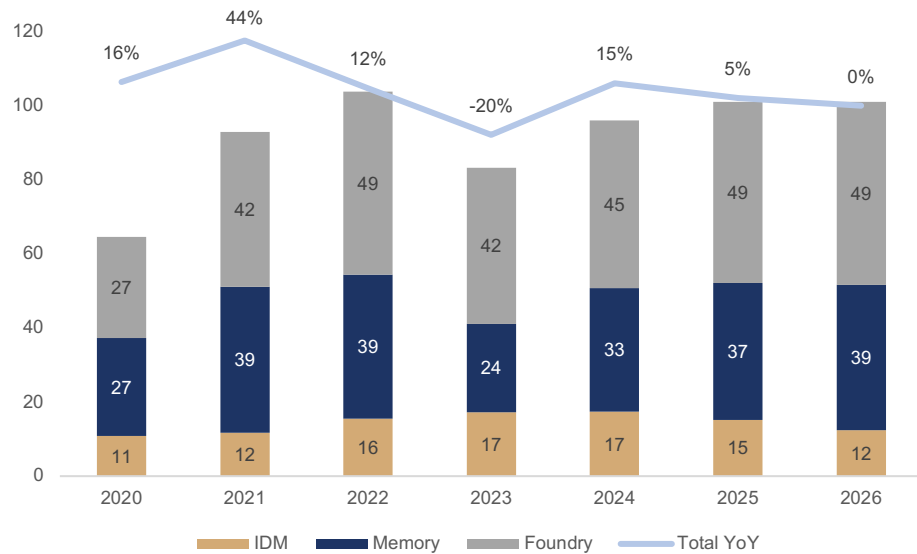
Figure 33: CoWoS demand forecast (kwpm)



Sources: Company data, GF Securities (Hong Kong) Brokerage

Figure 34: CoWoS supply chain


Sources: Company data, GF Securities (Hong Kong) Brokerage

Figure 35: Global WFE forecast (USD bn)


Sources: Bloomberg, Semi, GF Securities (Hong Kong) Brokerage

Upstream tech – Optical

2026 800G demand better than expected

Demand for 1.6T in 2026 better than expected. We update that the total shipment of 1.6T will reach 6mn in 2026, mainly driven by Google's better than expected demand. We expect that the upward revision of the 1.6T is favorable for overseas/local optical module players like **COHR/LITE** and switch players like **CLS**.

Shipment of TPU revised up in 2025, driven by better demand and delay of TPUv7. TPUv7 has strong momentum in 2026. Other than TPU, we expect that MTIA v1/v1.5 version to be ~0.5m in 2026, less than market's positive expectation of over 1m. Our optimistic assumption for Meta's 800G optical module will be 9m in 2026. Meta's Minerva's scale out L1 may use optical module or AEC. If the L1 all use optical module, the ASIC to optical module ratio will be 1:7. Considering GPU and traditional data center upgradation, we expect that the optimistic outlook for Meta's 800G optical modules in total will be 9m.

Figure 36: optical module TAM estimate

Revenue (USD mn)	2018	2019	2020	2021	2022	2023	2024	2025E	2026E
400G	50	127	554	1,291	1,470	1,082	3,045	3,920	2,314
800G	-	-	-	13	95	1,500	6,500	11,460	15,921
1.6T	-	-	-	-	-	-	-	1,800	6,200
3.2T	-	-	-	-	-	-	-	-	-
SUM	50	127	554	1,303	1,565	2,582	9,545	17,180	24,435
Shipment (m units)	2018	2019	2020	2021	2022	2023	2024	2025E	2026E
400G	0.0	0.2	1.0	2.9	3.8	3.3	10.2	14.0	8.9
800G	-	-	-	0.0	0.1	2.0	10.0	22.0	36.6
1.6T	-	-	-	-	-	-	0.2	1.5	6.2
3.2T	-	-	-	-	-	-	-	-	0.2
SUM	0.0	0.2	1.0	2.9	3.9	5.3	20.4	37.5	51.9
YoY (%)									
400G		275%	533%	207%	29%	-14%	212%	38%	-36%
800G					900%	1900%	400%	120%	66%
1.6T								650%	313%
3.2T									
ASP(USD)	2018	2019	2020	2021	2022	2023	2024	2025E	2026E
400G	1,262	845	583	442	391	333	300	280	260
800G	-	-	-	1,265	946	750	650	520	435
1.6T	-	-	-	-	-	-	-	1,200	1,000
YoY (%)									
400G		-33%	-31%	-24%	-12%	-15%	-10%	-7%	-7%
800G					-25%	-21%	-13%	-20%	-16%
1.6T									-17%

Source: GF Securities (Hong Kong) Brokerage

Figure 37: 800G/1.6T optical module demand forecast by customer

Shipment(m)	2018	2019	2020	2021	2022	2023	2024E	2025E	2026E
800G				0.0	0.1	2.0	10.0	22.0	36.6
Google				0.0	0.1	1.0	3.5	4.0	6.0
AWS								4.0	6.0
Meta						0.2		4.0	9.0
NVDA						0.8	6.0	3.0	3.0
MSFT								2.0	4.0
Alibaba								0.0	0.8
Tencent								0.0	0.8
Huawei								0.5	3.0
Others							0.5	4.5	4.0
1600G							0.2	1.5	6.2
Google							0.1	0.5	2.0
AWS									0.1
Meta									0.5
NVDA							0.1	1.0	3.0
MSFT									0.1
Alibaba									-
Tencent									-
Huawei									-
Others									0.5

Source: GF Securities (Hong Kong) Brokerage

Figure 38: GPU/ASIC shipment forecast

	2024	2025E	2026E	2027E
GPGPU (Units m)				
NVIDIA	3.9	5.9	7.7	7.2
AMD	0.5	0.6	0.8	0.9
Others	0.7	1.3	1.9	3.2
Total	5.1	7.9	10.4	11.3
YoY	160%	55%	32%	9%
ASIC (Units m)				
Google	2.8	3.1	3.2	4.2
AWS	0.9	1.8	2.4	3.2
Meta	0.2	0.3	0.5	0.8
MSFT	0.1	0.0	0.1	0.2
Other ASICs	0.3	0.8	1.0	2.0
Total	4.2	6.0	7.2	10.4
YoY	125%	41%	19%	45%
Market (USD bn)				
NVIDIA	88	156	199	231
Total GPGPU	98	173	225	270
Total ASIC	20	28	41	78
Total Market	118	200	266	348
YoY	181%	69%	33%	31%

Source: Company data, GF Securities (Hong Kong) Brokerage



Risks

1) Alternative technologies; 2) AI demand and capex slowdown; 3) IT demand slowdown; 4) product delays

Rating definitions

Benchmark: Hang Seng Index (Hong Kong)

Company ratings

Buy	Stock expected to outperform benchmark by more than 15%
Accumulate	Stock expected to outperform benchmark by more than 5% but not more than 15%
Hold	Expected stock relative performance ranges between -5% and 5%
Underperform	Stock expected to underperform benchmark by more than 5%

Sector ratings

Positive	Sector expected to outperform benchmark by more than 10%
Hold	Expected sector relative performance ranges between -10% and 10%
Cautious	Sector expected to underperform benchmark by more than 10%

Disclaimer

This report has been prepared by GF Securities (Hong Kong) Brokerage Limited ("**GF Securities (Hong Kong) Brokerage**"). According to the laws, regulations and regulatory requirements in different countries and regions, this report is distributed by GF Securities (Hong Kong) Brokerage with relevant legal and compliant operation qualifications in these countries and regions.

GF Securities (Hong Kong) Brokerage is licensed by the Securities and Futures Commission of Hong Kong ("**SFC**") to conduct Type 4 Regulated Activity "Advising on Securities". It is regulated by the SFC, and is responsible for the distribution of this report in Hong Kong. Information about the qualifications of the research analyst(s) who is(are) the author(s) of this report as licensed by the SFC are disclosed in the section where research analyst names are shown.

The research analyst(s) primarily responsible for the content of this report, in whole or in part, certifies that with respect to the company or relevant securities that the analyst(s) covered in this report: (1) all of the views expressed accurately reflect his or her personal views on the company or relevant securities mentioned herein; and (2) no part of his or her remuneration was, is, or will be, directly or indirectly, in connection with his or her specific recommendations or views expressed in this report.

This report is published solely for information purpose and does not constitute an offer to buy or Hold any securities or a solicitation of an offer to buy, or recommendation for investment in, any securities.

The securities mentioned in this report may not be allowed to be sold in certain jurisdictions. No action has been taken to permit the distribution of the research report to any person in any jurisdiction that the circulation or distribution of such research report is unlawful. This report is distributed solely to clients or designated institutions authorized by GF Securities (Hong Kong) Brokerage, and is not distributed publicly. It is distributed to certain clients based on the conclusion that they are able to assess investment risks independently, execute investment decisions independently and assume corresponding risks independently.

This report has been issued and based on information obtained from sources generally available to the public and believed by the research analyst(s) to be reliable but which has not been independently verified. No representation or warranty, either express or implied, is made by GF Securities (Hong Kong) Brokerage as to their accuracy and completeness of the information contained in this report. GF Securities (Hong Kong) Brokerage accepts no liability for all loss arising from the use of the materials presented in this report, unless is excluded by applicable laws or regulations. Please be aware of the fact that investments involve risks and the price of securities may be fluctuated and therefore return may be varied, past results do not guarantee future performance. Any recommendation contained in this report does not have regard to the specific investment objectives, financial situation and the particular needs of any individuals. This report is not to be taken in substitution for the exercise of judgment by respective recipients of this report, where necessary, recipients should obtain professional advice before making investment decisions.

GF Securities (Hong Kong) Brokerage may have issued, and may in the future issue, other communications that are inconsistent with, and reach different conclusions from, the information presented in the research report. The points of view, opinions and analytical methods adopted in the research report are solely expressed by the analysts but not that of GF Securities (Hong Kong) Brokerage or its affiliates. The information, opinions and forecasts presented in the research report are the current opinions of the analysts as of the date appearing on this material only which may subject to change at any time without notice. The salesperson, dealer or other professionals of GF Securities (Hong Kong) Brokerage may deliver opposite points of view to their clients and the proprietary trading division with respect to market commentary or dealing strategy either in writing or verbally. The proprietary trading division of GF Securities (Hong Kong) Brokerage may have different investment decision which may be contrary to the opinions expressed in the research report. GF Securities (Hong Kong) Brokerage or its affiliates or respective directors, officers, analysts and employees related to research report business may have rights and interests in securities mentioned in the research report. Recipients should be aware of relevant disclosure of interest (if any) when reading this report.

GF Securities (Hong Kong) Brokerage and its affiliates may be seeking or building business relationships with company(ies) mentioned in this report. Therefore, investors should consider the impact on the independence of this report by GF Securities (Hong Kong) Brokerage and its affiliates due to potential conflicts of interests. Investors should not make any investment decisions based solely on the contents of this report. Investors should make their own investment decisions and bear their own risk. No written or verbal commitment of sharing gains or losses from securities investments in any form shall be effective.

Please be sure to read the disclosures at the end of this report carefully.

This report may contain and/or describe/present factual historical information on prices of Futures contracts (the “information”). Please note that this information is solely for the purpose of forming part of the argument/grounds/evidence in our research methodology/analysis to support our conclusion on our view of the relevant industry/company mentioned. It does not, by any means (express or implied) to be associated with or constituted as SFC Type 5 Regulated Activity (Advising on futures contracts).

Disclosure of Interests

- (1) The research analyst(s) and his/her associate has not served as an officer of the company(ies) mentioned in this report, and does not have any financial interests in the company(ies) mentioned in this report.
- (2) GF Securities (Hong Kong) Brokerage and/or its affiliates does not have any financial interests and has not invested in interests aggregate to an amount equal to or more than (i) 1% of the market capitalization; or (ii) 1% of the issued share capital, or issued units, in the company(ies) mentioned in this report.
- (3) GF Securities (Hong Kong) Brokerage and/or its affiliates does not have any market making activities, and has not employed any individual(s) serving as officer(s) of the company(ies) mentioned in this report.
- (4) GF Securities (Hong Kong) Brokerage and/or its affiliates does not have any investment banking relationship in Hong Kong with the company(ies) mentioned in this report in the past 12 months.

Copyright © GF Securities (Hong Kong) Brokerage

Without the prior written consent obtained from GF Securities (Hong Kong) Brokerage, any part of the materials contained herein should not (i) in any forms be copied or reproduced or (ii) be re-disseminated.

Hong Kong	
Company:	GF Securities (Hong Kong) Brokerage Limited
Address:	27/F, GF Tower, 81 Lockhart Road, Wan Chai, Hong Kong
Telephone:	(852) 37191111
Email:	evanlee@gfgroup.com.hk